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The complex nature of regional knowledge production: Evidence on European regions

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ABSTRACT

Knowledge creation is widely considered as the central driver for innovation, and accordingly, for creating competitive advantage. However, most measurement approaches have so far mainly focused on the quantitative dimension of knowledge creation, neglecting that not all knowledge has the same value (Balland and Rigby, 2017). The notion of knowledge complexity has come into use in this context just recently as an attempt to measure the quality of knowledge in terms of its uniqueness and its replicability. The central underlying assumption is that more complex knowledge is more difficult to be replicated, and therefore provides a higher competitive advantage for firms, or at an aggregated level, regions and countries. The objective of this study is to advance and apply measures for regional knowledge complexity to a set of European regions, and to highlight its potential in a regional policy context. This is done by, *first*, characterising the spatial distribution of complex knowledge in Europe and its dynamics in recent years, *second*, establishing that knowledge complexity is associated with future regional economic growth, and *third*, illustrating the usefulness of the measures by means of some policy relevant example applications. We proxy the production of complex knowledge with a regional knowledge complexity index (KCI) that is based on regional patent data of European metropolitan regions from current EU and EFTA member countries. The results are promising as the regional KCI unveils knowledge creation patterns not observed by conventional measures. Moreover, regional complexity measures can be easily combined with relatedness metrics to support policy makers in a smart specialisation context.

1. Introduction

The ability to create and adapt new knowledge as a pre-requisite for successful innovation is widely considered as an important driver for the competitiveness of firms, and accordingly the regions where these firms are located (see e.g. Malecki, 2014). Therefore, the development and application of indicators on knowledge production has become one of the central concerns not only in a scientific context per se, but increasingly as a major instrument to orientate and evaluate research and innovation policies (see e.g. Godin, 2003; Katz, 2006), often referred to under the notion of policy learning and evidence-based policy making (Borrás and Laatsit, 2019; Sanderson, 2002). In this context, many scholars point to the growth of the 'indicators landscape' in terms of quantity and data availability, which is at the same time characterised by more limitations in times of increased complexity of innovation processes and systems (see e.g. Freeman and Soete, 2009). To remedy this, specific interest has been shifted to the notion of *positioning indicators*, pointing to the emerging needs of re-designing indicators for knowledge production and innovation in terms of wider changes in the research and innovation system (Lepori et al., 2008).

One weakness that we can specifically observe of indicators on knowledge production – with knowledge production considered as the main underlying driver for innovation – is its focus on *quantity*. In other words, common indicators measuring knowledge production activities, often derived from patenting information, focus on pure quantitative counts of some knowledge outputs, relating such outputs to innovative outcomes. This is also common practice for most policy-oriented indicator systems, such as the European innovation scoreboard (Hollanders et al., 2009). Accordingly, in such works it is implicitly assumed that all knowledge has the same value, i.e. the quality of knowledge is often neglected (Balland and Rigby, 2017). This seems a major weakness, in particular when comparing knowledge production activities and interactions at the regional or country level, which has gained increasing interest in the recent past (see e.g. Scherngell, 2013). Especially the

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regional level has become a focus as a relevant unit of intervention for policy makers (Asheim et al., 2011), also reflected by the growing relevance of concepts for regional smart specialisation strategies (Foray et al., 2009). From the economics of knowledge literature (Foray, 2004), we know very well that knowledge or technologies easily to be imitated and diffused in geographical space offer less opportunity for sustainable innovation success. In contrast, the possession of knowledge that is hard to replicate and to move is considered to constitute a major competitive advantage for firms, and regions where such firms are located. Accordingly, rather than counting the number of knowledge inputs and outputs it may be more important to assess the quality of knowledge produced, in particular from a research and innovation policy, or more specifically from a smart specialisation perspective (Balland et al., 2019).

Against this background, the notion of *knowledge complexity* has come into play more prominently in the recent past (see e.g. Balland and Rigby, 2017). Within this conception, the complex nature of knowledge is associated with its value and quality in terms of accessibility and mobility in (geographical) space, with a higher complexity reflecting increasing quality but decreasing accessibility for others due to its higher degree of 'tacitness'[1] and, accordingly, spatial 'stickiness'. In essence, the approach is derived from the work presented by Hidalgo and Hausmann (2009) who introduced the complexity notion to grasp the ability of countries to export non-ubiquitous product groups. This was transferred to the concept of knowledge complexity, and to the regional level (Balland and Rigby, 2017; Ivanova et al., 2017; Pintar and Scherngell, 2018) as an equivalent in terms of the knowledge domains *regions* are capable of. From the perspective of complexity and systems theory (Fleming and Sorenson, 2001; Kauffman, 1993; Simon, 1962), the complexity of knowledge can be well related to the variety of differing knowledge components it contains, and the interdependencies of those components.

In this context, the effects and importance of R&D collaboration networks and their increasing importance for knowledge production represent an important conceptual element of the discussion and should not be missed (Fritsch and Franke, 2004; Scherngell, 2013; van der Wouden and Rigby, 2019). In particular it is stressed that firms are not isolated from their environment, but embedded and inter-linked with their knowledge production activities in their regional surrounding exploiting localised capabilities (Maskell and Malmberg, 1999; Storper, 1997) comprised by the local or regional infrastructure, institutions, and endowments of resources, knowledge and skills (local buzz), but also the global economy (global pipelines) (Bathelt et al., 2004), enabling innovating actors to tap external knowledge sources (Breschi and Lenzi, 2016).

With its focus on knowledge complexity, this study lies in the vein of the research stream exploring knowledge production processes and their dynamics, but takes a spatial economics and regional policy perspective in its conceptualisation and application. The objective is to advance measures for regional knowledge complexity, and to apply them to a set of European regions, highlighting its potential in a regional policy context. In doing so, we define a regional knowledge complexity index (KCI) that intends to capture the complex nature of a region's knowledge production based on patent figures. The KCI relies on the diversity of a region's patent portfolio in terms of the technological fields the patents are applied in, and its ubiquity within a network of technologies recorded in patents. Note that when measuring a region's patenting activity, we weight the importance of a patent in terms of its ability to influence other knowledge domains with the number of citations a patent has received from patents located in other regions. This weighting scheme is utilised to account for knowledge 'exporting' capabilities via patent citations and thus make the KCI conceptually more comparable to the economic complexity index by Hidalgo and Hausmann (2009). Moreover, with the usage of patent citations, we also proxy knowledge 'exports' through inter-regional R&D linkages of regions to a certain extent. Lastly, extensive literature on the usage of patents suggests that patent citations received relate to the social, technological and economic value of patents (see e.g. Trajtenberg, 1990).

The study departs from previous related research in at least three major aspects: *First*, we present an original spatial pattern of regional knowledge complexity for a set of European regions, highly relevant in a European smart specialisation and policy context, in particular in terms of the debate on the limited operationalisation of smart specialisation policies. From the perspective of a specific region, diversification into more complex technologies may be an attractive, though difficult, element of such policies, but this lacks sound empirical measurement approaches so far (Balland et al., 2019). *Second*, we advance existing measures for regional knowledge complexity by weighting the patent portfolio of a region with received citations from patents located in other regions. By doing this, we adjust the raw counts of regional patents by approximate patent value and their relevance for knowledge production activities in other regions. *Third*, we apply the regional knowledge complexity measures to a new and more meaningful set of European regions. Most works investigating regional knowledge production rely on standard administrative classifications (mostly NUTS-2) that might artificially intersect agglomerations of knowledge creation (cities) leading to problematic interpretations in a spatial context (Lepori et al., 2019). If a considerable share of residents in areas close to or around large cities commute to work, economic activity in these commuter belts should be counted towards the main city, essentially creating functional areas. The so-called *metropolitan regions*[2] defined by EUROSTAT (2019) aim to do that. In order to best measure (high-quality) knowledge production at the location it actually occurs, we utilise this new set of European city-regions. Moreover, (big) city-regions seem to be much more prolific and efficient in high-quality knowledge production (Balland et al., 2020).

The remainder of this study is organised as follows. Section 2 discusses the conceptual foundations of knowledge complexity and its relation to geography, inter-regional linkages and relatedness, before outlining operationalisations. Section 3 introduces employed data and Section 4 discusses the results. The last section closes with a discussion, conclusions and a short outlook on a future research agenda.

2. The concept of knowledge complexity and its operationalisations

2.1. Conceptional considerations

The notion of knowledge complexity has been highlighted recently in innovation studies when explaining the spatial distribution of innovative activities (e.g. Balland and Rigby, 2017). In essence, it is argued that traditional studies that try to measure knowledge production lack a consideration of the quality, 'tacitness' and value of knowledge, summarised under the heading of knowledge complexity. The difficulty to conceptualise these dimensions in a rigorous manner, and operationalise

[1] The literature differentiates codified from tacit knowledge (see e.g. Gertler, 2003). The first type corresponds to knowledge that is written down and made explicit, whereas the latter is understood as skills, routines and ideas that are inherent to economic agents and not easily communicated (Polanyi, 1966). The endowment with tacit knowledge forms a central basis for innovation activities and its uneven distribution is a key determinant of the concentration of innovation output in space (Asheim and Gertler, 2005; Maskell and Malmberg, 1999; Pavitt, 2002). Tacit knowledge is spatially sticky because it is by definition not codifiable without difficulty and because it is often very dependent on the social and institutional context in which it was produced (Gertler, 2003; Polanyi, 1966). This relative immobility of tacit knowledge strengthens the competitive position of highly innovative regions (Asheim and Gertler, 2005).

[2] See Section 3 for details.

them in a robust way is quite obvious, given the – a priori – latent and diffuse nature of knowledge, and even more its value.

Knowledge complexity can be understood as a multi-dimensional measurement concept to capture these dimensions of knowledge (Broekel, 2017). While there have been notable attempts at the firm level (see e.g. Singh, 1997), there are only a few and very recent attempts to conceptualise and define knowledge complexity for countries or regions. These attempts mostly build on the influential work of Hidalgo and Hausmann (2009). Their concept of economic complexity grasps the ability of countries to export non-ubiquitous product groups, which can only be traded by relatively few countries. The fact that a country is able to export such sophisticated products competitively should signal the existence of a large set of necessary latent (technological) capabilities which in turn should constitute a competitive advantage. Indeed, economic complexity has been shown to be superior in predicting future economic growth of a country to other indicators such as education and institutional quality (Hausmann et al., 2011). This fundamental concept of economic complexity can be very well translated to (regional) knowledge production and has recently been applied in the context of knowledge complexity.

In this study, we follow this recent research direction of employing the approach of Hidalgo and Hausmann (2009) to capture the complexity of knowledge of spatial entities (countries or regions), using their technological patent portfolio (patents of a specific technological domain applied for in a specific region), and citations to these patents from extra-regional inventors as proxy for their knowledge 'exporting' activities and R&D linkages. Then we combine – in the same way as done with exports for the economic complexity index – the diversity and ubiquity of the citation-adjusted patent portfolio of a country/region in a knowledge complexity index for countries or regions (see e.g. Antonelli et al., 2017; Balland et al., 2019; Balland and Rigby, 2017; Ivanova et al., 2017; Whittle, 2017). It is argued that the knowledge complexity index constitutes an advancement of the economic complexity index itself, since the latter is heavily used to draw conclusions on technological capabilities of countries, but neglects an explicit definition of the manufacturing capabilities and their respective knowledge bases (Ivanova et al., 2017). In terms of interpretation, the knowledge complexity index proposed by Balland and Rigby (2017) of countries or regions is understood as their ability to create and sustain knowledge bases that are non-ubiquitous in the system.

Balland and Rigby (2017) show a strong spatial concentration and considerable path dependency in the evolution of complex knowledge among U.S. city regions. Moreover, they also find that complex knowledge is indeed spatially more 'sticky' than more ubiquitous knowledge. Other recent papers show a positive impact of knowledge complexity on subsequent knowledge production (Antonelli et al., 2017; Balland et al., 2019), evidencing again the path-dependent nature of knowledge production.

From an economics of knowledge perspective, it is worth stressing that regional actors of knowledge production create new knowledge by recombining knowledge sets or ideas that are present in the region or tapped externally through R&D collaborations (Bathelt et al., 2004; Fleming and Sorenson, 2001). While both local and non-local research collaborations are important for innovative firms to produce new knowledge (Esposito and Rigby, 2019), van der Wouden (2019) shows that more complex knowledge is developed and sourced more often through local collaborations while less complex knowledge depends less on the geographic proximity of collaborators. This tendency for mainly local collaborations of complex knowledge adds to literature identifying the spatial immobility of complex knowledge, which further strengthens the expectation of sustained economic benefits to regions that produce such knowledge.

This mentioned research on knowledge complexity is specifically relevant in the context of the regional smart specialisation debate that has gained pace in recent years (Foray et al., 2009, among others). Smart specialisation is about the individual diversification of regions into promising, usually related, knowledge domains leveraging regional strengths and opportunities. In this sense, the concept of knowledge complexity, and accordingly its empirical implementation can have a significant added value for the operationalisation of smart specialisation strategies that is rather limited so far (Balland et al., 2019).

As is well established in the literature, diversification into activities that are related to current expertise, skills and knowledge of individuals or economic agents located in regions is considerably easier and more fruitful (see e.g. Boschma and Frenken, 2011; Essletzbichler, 2015; Frenken et al., 2007; Hidalgo et al., 2018). The diversification of regions into related technologies is widely assumed to be conducive for regional innovation (see e.g. Essletzbichler, 2015; Kogler et al., 2017; Rigby, 2015), and accordingly, regional economic activity, but whether diversifying into more complex domains is crucial is only recently being analysed (Balland et al., 2019; Whittle, 2017).

2.2. Operationalisations of knowledge complexity

Departing from these conceptual elaborations, we aim to define a measure for knowledge complexity applied at the regional level of analysis. Building on the approach of Hidalgo and Hausmann (2009), a region's knowledge complexity is understood as a function of its diversity in terms of different technologies used to produce and transfer regional knowledge to other regions, and the ubiquity of these technologies, i.e. how many other regions are capable of producing and 'exporting' knowledge related to a specific technological field. Accordingly, the knowledge complexity of regions is based on the region-by-technology network matrix, representing the technological portfolio of all regions as it connects each spatial entity $i = (1 \ldots N)$ with the technological fields $k = (1 \ldots K)$ in which it is specialised in. Similar to previous literature we use the concept of Revealed Technological Advantage (RTA) by Soete (1987) to find apparent specialisations of regions in technologies for the time period given by t (subscript t omitted for clarity purposes).

$$RTA_{ik} = \frac{X_{ik}}{\sum_k X_{ik}} \Big/ \frac{\sum_i X_{ik}}{\sum_i \sum_k X_{ik}} \tag{1}$$

The RTA of a region in a specific technological field is the ratio between the share of the regions' knowledge production in this field and the share of the same technological field in the whole sample. X_{ik} can be any proxy of knowledge production of region i in technology k. We use citation-weighted patent applications to account not only for knowledge production, but whether this new knowledge is taken up by other regions. A value larger than one signals a relative regional specialisation in the specific field. Consequently, we define the matrix M as

$$M_{ik} = \begin{cases} 1 \ if \ RTA_{ik} > 1 \\ 0 \ if \ RTA_{ik} \leq 1, \end{cases} \tag{2}$$

i.e. elements are set to 1 if a region is specialised in a certain technology, and to zero otherwise. M can – from a graph theoretic perspective – also be described as a bipartite graph with two distinct sets of nodes (the N regions and K technological fields) where only nodes of different types can be connected. Region i is connected to field k in the European knowledge production network if, and only if, $M_{ik} = 1$. The diversity in knowledge production of region i is then simply given by its degree centrality, $d_i = \sum_k M_{ik}$. Analogously, the ubiquity of k is equal to its degree centrality, $u_k = \sum_i M_{ik}$. Hidalgo and Hausmann (2009) introduced the so-called *Method of Reflections* in order to infer the complexity of countries (and products) from the network of global exports of products. Translated to our notation and applied to knowledge production, this iterative, self-referential algorithm (see Eq. (3)) takes regional diversification ($d_i = d_i^0$) and the ubiquity of technological fields ($u_k = u_k^0$) and then recursively refines these variables with n iterations to yield estimates of regional and technological complexity ($d_i^n; u_k^n$).

$$\begin{cases} d_i^n = \dfrac{1}{d_i^0}\sum_k M_{ik}\ u_k^{(n-1)} \\ \\ u_k^n = \dfrac{1}{u_k^0}\sum_i M_{ik}\ d_i^{(n-1)} \end{cases} \tag{3}$$

In other words, this algorithm produces generalised measures of diversification and ubiquity where each iteration uses information from previous iterations to yield a finer estimate of regional and technological complexity, respectively. Each even iteration of d_i^n is a finer estimate of regional knowledge complexity, calculated as the average ubiquity of technological fields (at iteration $(n-1)$) in which this region is specialised in. Analogously, each uneven iteration of u_k^n produces a better estimate of technological complexity as the average diversification of regions (at iteration $(n-1)$) that are able to produce knowledge in that particular field (for further details see Caldarelli et al., 2012; Hidalgo and Hausmann, 2009; Mariani et al., 2015).

It has been shown that it is useful to reformulate the *Method of Reflections* as a fixpoint problem which can be solved analytically without the need of an iterative approach (Caldarelli et al., 2012). Finding the knowledge complexity of regions then corresponds to finding the eigenvector $\tilde{m}^{[2]}$ associated with the second largest eigenvalue of matrix $\tilde{M} = \widehat{M}\ \widehat{M}^{'}$, where $\widehat{M}$ and $\widehat{M}^{'}$ are the row-standardisations of M and its transpose. This row-stochastic square matrix $\tilde{M}$ with regions in both rows and columns shows weighted technological similarities between regions. Elements along the main diagonal can be interpreted as the average rareness of technological fields in which the row and column region has revealed technological advantage. Off-diagonal elements of matrix $\tilde{M}$ represent the rareness of common technological classes of the row and column region, averaged over all fields the row region is specialised in. We adopt this approach and define the KCI as equivalent to $\tilde{m}^{[2]}$.[3] A knowledge complexity index of technology classes (TCI) can be obtained by reversing the order of matrix multiplication; $\tilde{T} = \widehat{M}^{'}\widehat{M}$, and analogously solving for the eigenvector $\tilde{t}^{[2]}$ associated with the second largest eigenvalue of matrix $\tilde{T}$.

3. Data

Following related literature, we use patent data to proxy regional knowledge production, and citations to these patents to infer the flow of this knowledge to agents in other spatial entities (see Fischer et al., 2006; Jaffe et al., 1993). Specifically, we retrieve patent applications to the European Patent Office (EPO) by EU and EFTA inventors between 2005 and 2014[4] from the OECD REGPAT[5] database, which offers regionalised patent data. Patents are allocated to NUTS-3 regions in the REGPAT database, where patents are attributed to regions by inventor residence. We map patents located in these NUTS-3 regions to metropolitan regions as defined by EUROSTAT[6] and remove (fractional) patents that are located in peripheral regions according to this classification. Because most knowledge production activity happens in or around big agglomerations, this only removes 22% of patents from the dataset. Moreover, these metropolitan regions are expected to more realistically represent agglomerations of knowledge creation than standard NUTS regions because they minimise artificial intersections of urban agglomerations by NUTS borders. The patent office classifies patents according to the International Patent Classification (IPC). For our analysis we use the technological classification proposed by Schmoch (2008) which maps IPC classes onto 35 fields. This classification aims to strike a balance between homogeneity in terms of class sizes and technological homogeneity within each technological field and has been used in related literature with comparable datasets (Balland et al., 2019; Whittle, 2017).

It is recognised that the use of raw patent counts as measures of innovative output can be problematic because patents' value and importance vary extremely (Jaffe and Trajtenberg, 2002; Trajtenberg, 1990; Trajtenberg et al., 1997). To remedy this, scholars have analysed the relation of certain additional information[7] that is contained in patent documents with the social and economic value of patents (see e.g. Bakker, 2017; Bessen, 2008; Fischer and Leidinger, 2014; Hall et al., 2005; Lanjouw et al., 1998; Lerner, 1994). As mentioned in previous sections, we weight patent application counts by their citations received by subsequent patent applications in order to capture the knowledge 'exporting' capability of regions. In general, such patent forward citations have been shown to be related with the technological, social as well as economic importance of patents (Gambardella et al., 2008; Hall et al., 2005; Harhoff et al., 2003; Trajtenberg, 1990).

We complement our dataset with patent forward citation counts from the OECD Citations database, February 2019 (see Webb et al., 2005 for a description). Citations received[8] are counted over a time period of three years after publication of the patent by the EPO. They take into account citations of patent equivalents, i.e. patents at national patent offices that protect the same invention as the EPO patent (see Webb et al., 2005). Publication at the EPO typically occurs 18 months after filing. Thus, in order to not suffer from truncation, we can only take into

[3] Currently, there is an active ongoing discussion on the best way to extract information about the (economic) complexity of spatial units from the bipartite network that connects these units to products or knowledge subsets that they produce. In the context of economic complexity of nations, Tacchella et al. (2012) introduced a non-linear iterative algorithm that builds on a similar theoretical basis as Hidalgo and Hausmann (2009) but emphasises that the complexity of products should be more strongly limited by the least fit country that is able to export it. This approach termed Fitness-Complexity Method was shown to produce fitness scores that are strongly associated with economic (Cristelli et al., 2015; Tacchella et al., 2012) and scientific (Cimini et al., 2014) competitiveness of countries. Subsequent studies analysed the mathematical properties (Wu et al., 2016) and conditions for convergence of this algorithm (Pugliese et al., 2016), which revealed some limitations. The convergence of the algorithm to non-zero values for nodes of the network depends highly on the specific network structure. This is expected to be even more problematic when considering the network of technological portfolios than that of trade (Pugliese et al., 2016). Moreover, the strong limiting factor of the least fit exporting country to the complexity of products can lead to an overemphasis on niche products or technological classes. Hence, knowledge subsets in which only very few regions are specialised in could be identified as complex (and consequently their producing regions), even though they are not inherently difficult to produce but only rare (Morrison et al., 2017). For these reasons we focus on the method described in detail above.

[4] The year of filing at the EPO is used to date the patents considered.

[5] The OECD REGPAT database derives from the European Patent Offce's (EPO) Worldwide Statistical Patent Database (PATSTAT, Spring 2019); and the OECD patent database. We use the most recent version of the database as of writing this article; OECD REGPAT database, July 2019.

[6] Metropolitan regions are combinations of NUTS-3 regions which are aggregated in a way to more realistically represent urban regions and to ideally come close to functional regions of cities, including commuter belts around a city. NUTS-3 regions between urban agglomerations of at least 250.000 inhabitants are classified as periphery. We use metropolitan regions based on the NUTS 2013 classification.See https://ec.europa.eu/eurostat/web/metropolitan-regions/background for details.

[7] These include but are not limited to citations received by subsequent patents (forward citations) and citations included in the patent document at hand (backward citations), the grant status of the patent, the fact that a patent was filed at several patent offices, the number of times the patent's protective right was renewed by the patentee and potential legal actions concerning the patent document. For an overview of usable information contained in patent documents see van Zeebroeck (2011).

[8] Here, we count citations coming from EPO patents as well as patents in the PCT route.

account patents that were filed until 2014.[9] We weight patent application counts with patent forward citations in the following way.

$$w_p = \frac{c_p + 1}{\mu_{tk} + 1} \tag{4}$$

Each patent application p receives a weight w_p that is dependent on the number of citations received by patent p from extra-metropolitan patents (c_p) in relation to the mean citation count of the corresponding patent cohort (μ_{tk}). It has been shown that a considerable number of patent citations come from either the same inventor/assignee or the same region (Breschi and Lissoni, 2009; Jaffe et al., 1993; Jaffe and de Rassenfosse, 2017). We remove intra-metropolitan patent citations[10] to reflect the theoretical argument of knowledge complexity as the capability to 'export' sophisticated knowledge. In order to not drop patents that received zero citations but rather only deem them less valuable, we add 1 to each the citation count and the corresponding cohort mean. We compare the number of forward citations received with the patent cohort in regards to year and technological field because it is well known that the propensity to cite varies widely between technologies (Lerner and Seru, 2017; Squicciarini et al., 2013).

The weight w_p of each patent can then be interpreted as the patent's importance relative to the average patent filed at the EPO in the same year and pertaining to the same broad technology group. Citation-weighted patents are fractionally counted,[11] i.e. equally split among their respective inventors' regions and technological classifications. Because the number of patents per region varies widely per year, with some regions counting almost no patents, we aggregate citation-weighted patents over a period of five years, as is done in many related studies (see e.g. Antonelli et al., 2017; Broekel, 2017). Due to the slow and path-dependent nature of technological knowledge production, it also makes theoretical sense to sum up patent applications of various years (Antonelli, 2009). Despite the aggregation of five years, many regions barely produce any patents. This is not unexpected considering the inclusion of Eastern and Southern European regions, which historically have recorded very few patents (Fischer et al., 2009). Even though we want to include as many regions as possible, a very small number of patents might adversely affect our empirical results due to biased RTA distributions (Cantwell and Vertova, 2004; Laursen, 2015). Moreover, excluding urban agglomerations that are apparently not very active in producing patents is not detrimental to identifying metropolitan regions that top the list in terms of complex knowledge production. Thus, we define a threshold set at 100 fractionally-counted patents per region, i.e. we exclude any region that did not produce at least 100 patents in the five-year period analysed.[12] After applying the threshold, we have a dataset of 247.038 / 247.098 unique patents that are linked to 200 / 211 European metropolitan regions and 35 distinct technological fields, respectively to the first and second time period. This data is used for any patent-related metrics in this study.

[9] The OECD Citation database February 2019 covers patent citations until 2018. Hence, in order to allow each patent the full three years after publication to be cited, 2014 is the latest year usable.

[10] It is our understanding that by removing all patent citations coming from the same metropolitan region, we exclude the vast majority of citations from the same inventor/assignee. About 34% of citations in our dataset were intra-metro citations.

[11] Note that many patents have several inventors listed who might live in different regions and that patents often contribute to various technological fields. Hence, double counting – in contrast to fractional counting - would likely inflate the number of patents considerably, which could lead to very different empirical results.

[12] By doing this, 73 metropolitan regions in the period 2005 to 2009 and 61 in the later period were excluded, mainly in Southern and Eastern Europe, see Fig. 2. Note, that patents are often allocated to more than one region and class which means that patents are not necessarily completely removed from the dataset once a region to which a patent belongs to is excluded. Alternative thresholds at 50,150 or 200 patents resulted in very similar results with rank correlations of regional KCI above 0.98.

Table 1
Top 25 metropolitan regions in terms of knowledge complexity, 2010-2014.

Metro code	Name	No. of patents	KCI	KCI rank	Div.
FR013M	Rennes	1442.7	100.0	1	6
SE001MC	Stockholm	5886.8	76.1	2	10
FR203M	Marseille	1751.8	72.3	3	9
FI002M	Tampere	1258.4	69.6	4	11
EE001MC	Tallinn	136.9	67.3	5	12
IE001MC	Dublin	970.1	67.0	6	13
FR205M	Nice	989.4	65.8	7	11
UK001MC	London	9615.3	64.4	8	14
ES001MC	Madrid	1675.2	63.5	9	11
SE006M	Uppsala	719.5	62.1	10	8
UK569M	Ipswich	307.9	61.5	11	11
NL005M	Eindhoven	7427.6	60.3	12	12
FI001MC	Helsinki	4142.0	58.8	13	11
UK005M	Bradford	160.4	57.7	14	12
FR001MC	Paris	15242.4	57.7	15	16
UK023M	Portsmouth	1764.7	57.2	16	13
SE003M	Malmö	2993.4	56.6	17	15
UK513M	Medway	105.6	55.1	18	7
FI003M	Turku	538.6	54.8	19	13
PL004M	Wroclaw	133.2	54.7	20	16
UK007M	Edinburgh	612.9	54.2	21	14
ES026M	A Coruña	117.7	53.1	22	12
HU001MC	Budapest	706.4	51.3	23	7
IE002M	Cork	352.6	50.4	24	15
BE002M	Antwerpen	1147.5	50.4	25	16

Notes: Patents are citation-adjusted and fractionally counted. KCI score is re-scaled to values between 100 and zero.

4. Results

In this section we turn to the empirical results where we first reflect on the overall ranking of the regional KCI as well as the dynamics of knowledge complexity in recent years and illustrate spatial patterns of the distribution of complex knowledge in Europe. Second, we present an econometric model that is intended to demonstrate a certain credibility and usefulness of the proposed knowledge complexity index beyond the information conveyed descriptively. Finally, we propose an identification framework related to (Balland et al., 2019) and identify regions' diversification potential in terms of high-value technological knowledge production.

4.1. Knowledge complexity of European metropolitan regions

In Table 1 we list the top 25 city-regions in terms of regional knowledge complexity (KCI) calculated for the most recent period analysed as defined in the previous section.

The results are interesting, as only Paris as part of the usual suspects showing the highest overall patenting activity (Paris, München, Stuttgart, etc.) makes it into the top of the KCI ranking. Instead, we find a mixture of small and large regions in terms of citation-adjusted patent count within the top 25 regions according to the knowledge complexity index. These regions receive a high complexity score because they are specialised in highly complex technologies (see Table A4 for the full ranking of technologies in terms of complexity). For example, notwithstanding their relatively low diversification, Rennes, Stockholm, and Marseille all specialise in the top 3 most complex technologies (among others) as identified by the network of European patent portfolios. Interestingly, there seems to be a positive but rather weak relation between regional knowledge complexity and the citation-adjusted quantity of patents produced as well as the overall (non-complex) diversification of knowledge production (see Table A3: $\rho \approx 0.29; 0.05$, respectively). While a positive correlation makes theoretical sense, a low correlation possibly indicates that the KCI captures information about

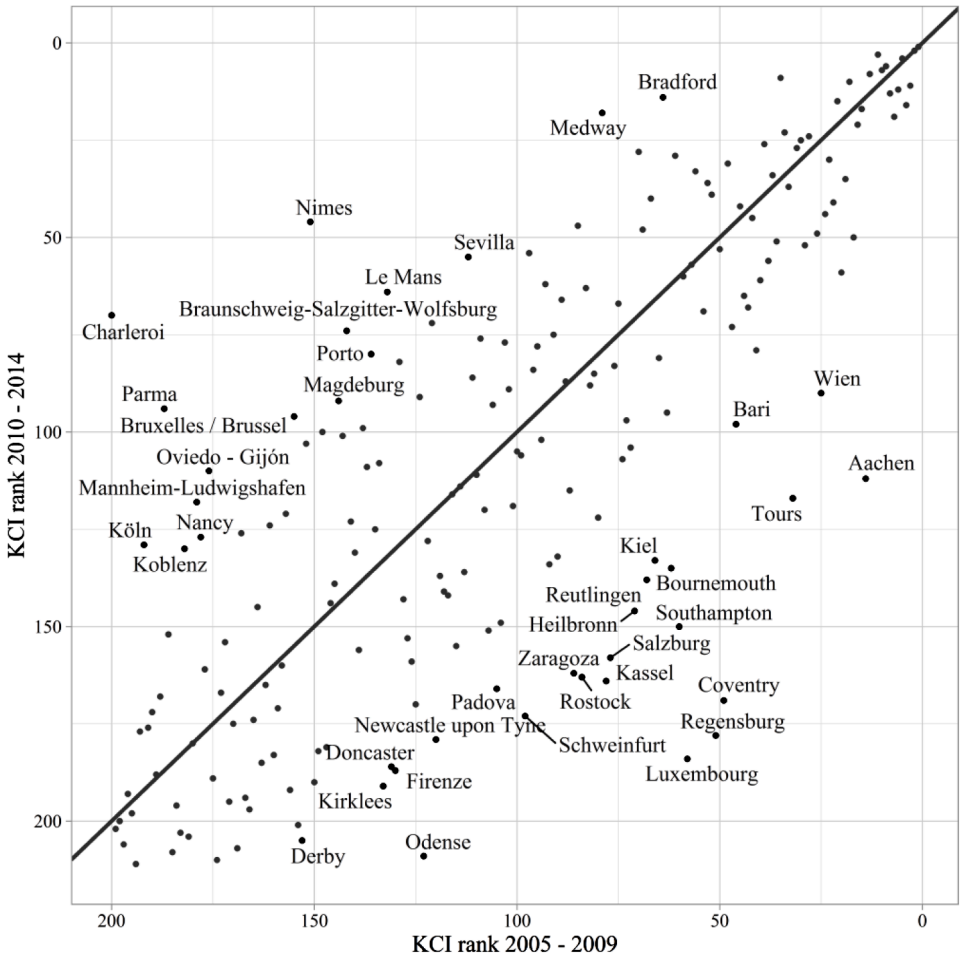


Fig. 1. Changes of KCI rankings.
Notes: Metropolitan region names are omitted when their ranking did not change at least 50 positions. Only regions that are included in both time periods are shown.

regional knowledge bases that is not captured by simply looking at patent activity or overall diversification of a region.[13]

In Fig. 1 we show changes of the regional KCI ranking between the two periods analysed. Overall, we see a certain rigidity of the regional ranking of knowledge complexity with some notable exceptions. The rank correlation of regional knowledge complexity between 2005-2009 and 2010-2014 is 0.77 and the average rank change is -5.9. While most regions stayed relatively close to their position in the previous period, we find that some regions managed to climb up the ladder quite drastically. For example, as seen in Fig. 1, Charleroi (Belgium) and Nimes (France) both diversified their patent portfolio into more complex technologies and jumped more than 100 ranks. In contrast, Regensburg (Germany), Luxembourg, and Coventry (UK) focused their knowledge production activities in less technological fields that are also less complex and dropped more than 100 ranks as a consequence. Bradford (UK) and Medway (UK) also represent interesting cases because they entered the top 20 from rank 64 and 79, respectively. Notably, this shift is due to a higher relevance of more complex technologies in their patent portfolio, and not so much to an increased diversification. Changes in the patent portfolios of city-regions can be due to a multitude of reasons that need to be analysed on a case-by-case basis. Still, some dynamics may be attributable to policy interventions, e.g. in the case of Charleroi where a 'Marshall Plan for Wallonia'[14] was introduced in 2005 to specifically increase innovation (among other targets).

Shifting attention towards the spatial distribution of complex knowledge in European regions (see Fig. 2) for the most recent period, it seems that there exists a clear spatial pattern where more (least) complex regions in terms of knowledge production are generally situated

[13] Current research employs a variety of operationalisations of a regional knowledge complexity index (see e.g. Mewes and Broekel, 2020; Rigby et al., 2019). As a robustness check, we also cross-check our results with a weighted average technological complexity score (WTCI) for each metropolitan region, as is used by Rigby et al. (2019). Here, an alternative knowledge complexity index for a region is computed as the average technological complexity index score (TCI), weighted by the share of each technology in the regional portfolio. The result shows a high robustness signalled by a pearson correlation coefficient of 0.94 for both sample time periods. In detail, we find that the top regions in terms of knowledge complexity are indeed very similar when comparing the two approaches. For example, Rennes and Stockholm also receive the highest scores with this alternative method. Moreover, only four out of the top 25 regions according to WTCI were outside of the top regions ranked by KCI. Results for the first period (2005-2009) were qualitatively similar. Given the established robustness and due to our conceptual focus on regions, which is inherent in the KCI conception per se, we prefer the approach established in the text.

[14] Charleroi is located in the French-speaking region of Wallonia in Belgium, a former heavy industry region that struggled massively with high unemployment and poverty rates in the last decades when these industries declined in Europe. The 'Marshall Plan for Wallonia' had a total funding of around EUR 1 billion for the period 2005 to 2009, included reductions in company taxation and aimed at promoting research and innovation (Van Haeperen et al., 2009). Follow-up projects broadened the scope to include cluster, circular economy and digital innovation policy and are now linked to the smart specialisation strategy of Wallonia (Larosse, 2017).

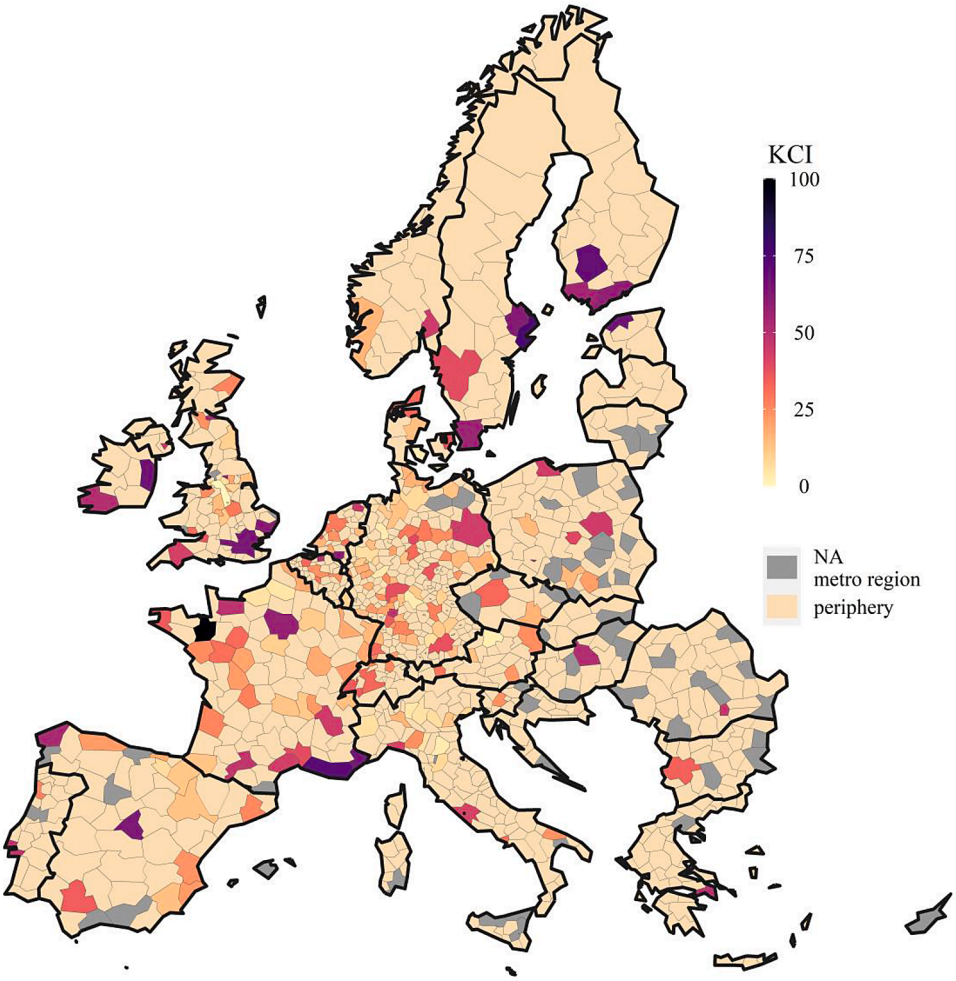


Fig. 2. Spatial distribution of regional knowledge complexity, 2010-2014.
Notes: KCI scores are re-scaled to values between 100 and zero. NA metro region refers to metropolitan regions that were removed from the dataset because they did not produce at least 100 patents in the five-year period analysed.

close to each other. This points to the existence of spatial spillovers, and a certain geographical logic in knowledge creation activities across the European territory. The most prominent example of this is an apparent cluster of complex knowledge production around Stockholm and Southern Finland. What is arguably most surprising is the identification of historically highly actively patenting regions in central Europe (Kogler et al., 2017) as relatively non-complex due to the specialisation of these regions in typical industrial technologies like *Machine tools, Mechanical elements* and others which are deemed non-complex. For example, Stuttgart (ranked 83th in terms of KCI) is historically very active in patenting. However, local inventors are relatively specialised in traditional industrial technologies[15] which are non-complex, according to the TCI.

Interestingly, while patent forward citations are used to weight patent production of metropolitan regions, the KCI cannot be simplified to the quantity of extra-regional citations received by these regions,[16] as can be easily seen when comparing Figs. 2 and A1. These results point to an interesting potential of the KCI in picking up novel patterns in European knowledge production from a complexity perspective.

As a complementary view, we also compare the knowledge complexity index with R&D expenditure per capita[17] as a yardstick indicator on whether the KCI mainly reflects financial knowledge production inputs (in addition to patents as output of knowledge production activities). Figure A 2 shows a low to moderate positive correlation between knowledge complexity and R&D expenditure p.c. in both time periods ($\rho \approx 0.28; 0.15$, respectively). It is clear that there should be a positive relation between knowledge production input and output. However, knowledge complexity is assumed to go beyond pure quantitative knowledge production inputs and outputs, but to estimate a specific quality of knowledge production. Given this, it is not surprising that R&D is much more associated with the number of patents produced ($\rho \approx 0.47$, in both periods) than with the KCI. Moreover, the fact that German city-regions typically have the highest R&D expenditure p.c. but score relatively low in knowledge complexity can further explain the low correlation, suggesting that high R&D expenditures in itself are not sufficient to produce complex knowledge.

[15] For example, inventors located in Stuttgart are highly active (i.e. Stuttgart has an RTA >1) in non-complex technological fields 31: *Mechanical elements*, 26: *Machine tools,* 27: *Engines, pumps, turbines,* among others.

[16] The correlation coefficient of knowledge complexity and the share of patents that fall in the top decentile in terms of citations received is only 0.13 in the most recent period (0.14 in period 1).

[17] We use intramural R&D expenditure (GERD) from EUROSTAT, measured in purchasing power standard (pps) per inhabitant at constant 2005 prices. The spatial unit of observation is NUTS2 2016. We translate this data to 2013 NUTS3 classification, assuming same R&D expenditure in NUTS3 regions within respective NUTS2 regions. Then we aggregate to get approximations for metropolitan regions.

4.2. Relating knowledge complexity to regional economic growth

In this section we present an econometric model that estimates the relation between gross regional product (GRP) growth (*GRP growth*) and knowledge production of European metropolitan regions. This addresses one of the main underlying debates on knowledge complexity, namely whether regions that are able to produce complex knowledge grow faster than other regions and become more productive in the long run. In this illustration, we focus on GRP growth, with regional *knowledge complexity (KCI)* – as derived in Section 2 and applied in the previous subsection – being our main variable of interest in explaining regional growth.

Consistent with common growth models, we include a set of control variables, including regional *initial GRP* to control for city size and possible convergence effects. Our main control variable *patents* denotes the number of citation-adjusted patent applications a region produced in the respective period of time. It is well established that knowledge production (often proxied by patent production) is a major determinant of economic growth (Antonelli, 2009; Asheim and Gertler, 2005; Feldman and Storper, 2018; Romer, 1990; Schumpeter, 1942). Following recent related literature (Balland et al., 2019; Boschma and Capone, 2015; Hidalgo et al., 2018; Petralia et al., 2017), we recognise that the proximity to technologies that are not yet in the technological portfolio of regions facilitates the diversification into these knowledge domains. We control for this ease of diversification with the inclusion of *relatedness density*. To distinguish the diversification in complex technologies from the broad scope of knowledge domains present in the region, we include *diversification*.

GRP data used in this study is sourced from ARDECO.[18] We measure relatedness following recent literature (Balland et al., 2019; Boschma et al., 2015; Rigby, 2015). Relatedness between two technological classes is a standardised[19] measure of the expected co-occurrence of these two technology classes on the same patent document. This is understood to proxy the distance between these technologies related to complementarity and ease of combination. Since our focus lies on regions and not necessarily on technologies, we want a metric that estimates the overall technological relatedness of knowledge production within a region. To that end we calculate the so-called *relatedness density*[20] for each region, which is derived from relatedness values between technologies present in that region. High values of this measure signal a considerable technological proximity of technologies present in the region with absent technologies.

Below we introduce four model specifications which are all estimated at the regional level with a dataset of 193[21] metropolitan regions in two time periods t (2005-2009, 2010-2014), matching the periods analysed previously in this study. Descriptive statistics are shown in Table A2. The baseline specification is given below in Eq. (5):

$$\begin{aligned} GRPgrowth_{i,t+1} \;=\; & \beta_1\, initialGRP_{i,t} + \beta_2\, KCI_{i,t} + \beta_3\, Patents_{i,t} \\ & + \beta_4\, RelatednessDensity_{i,t} + \beta_5\, Diversification_{i,t} \\ & + \beta_6\, const + \epsilon_{i,t} \end{aligned} \quad (5)$$

The dependent variable *GRP growth* is lead by one period, meaning the independent variables aim to explain GRP growth in the next period.[22] This is done for two reasons, first, theoretically we would not expect GRP effects of knowledge production to translate instantly (Antonelli, 2009; Feldman and Kogler, 2010), second, this approach might be able to reduce possible endogeneity issues in our regression framework. *Initial GRP* refers to regional GRP of the first year in the period and enters the model in log form. *KCI* is included normalised and *patents* also enters in log form. *Relatedness density* and *diversification* enter the model not transformed in any way. $\epsilon_{i,t}$ refers to the error term.

Turning to the results for the baseline model specification in Table 2, we find that the initial metropolitan GRP is significantly negatively associated with growth in the future. Considering standard growth

Table 2
Results of the econometric analysis.

Dep. variable: GRP growth				
	OLS	ML		
	baseline	SEM	SAR	SAC
	(1)	(2)	(3)	(4)
initial GRP (log)	-0.3720***	-0.3317**	-0.1506	-0.2560*
	(0.124)	(0.126)	(0.107)	(0.125)
Knowledge Complexity	0.2330***	0.2409**	0.3294***	0.2665***
	(0.087)	(0.077)	(0.072)	(0.076)
Patents (log)	0.3760***	0.3424**	0.1781*	0.2766*
	(0.100)	(0.109)	(0.088)	(0.109)
Relatedness Density	0.0400*	0.0274	0.0387*	0.0253
	(0.022)	(0.019)	(0.019)	(0.019)
Diversification	-0.0450	-0.0193	-0.0589	-0.0188
	(0.060)	(0.054)	(0.055)	(0.054)
Spatial error		0.8407***		0.8566***
		(0.101)		(0.099)
Spatial lag			0.5667***	0.6243**
			(0.130)	(0.216)
Constant	2.1800***	1.9770*	0.6813	1.3696
	(0.842)	(0.939)	(0.872)	(0.932)
Observations	386	386	386	386
R2	0.086			
Adjusted R2	0.074			
Log. Likelihood		-658.31	-662.81	-654.95

Notes: The dependent variable GRP growth is lead by one period. SEM, SAR and SAC refer to the spatial error model, the spatial autoregressive model and the spatial autoregressive combined model (see Lesage and Pace, 2009) which are estimated by Maximum Likelihood estimator (ML) using the splm R package. Heteroskedasticity-robust standard errors (clustered at the region) are shown in parenthesis for the baseline model. Coefficients are statistically significant at the *p<0.1; **p<0.05; ***p<0.01 level.

[18] ARDECO is the Annual Regional Database of the European Commission's Directorate General for Regional and Urban Policy. This helpful database is a collection of regional data from different sources but mainly from EUROSTAT. See https://ec.europa.eu/knowledge4policy/territorial/ardeco-database_en. GRP is deflated to 2015 values. In order to make GRP data comparable to the metropolitan patent data, we translate NUTS 3 data from ARDECO that comes in the NUTS 2016 classification first to NUTS 3 2013 classification and then aggregate this regional data to metropolitan regions.

[19] For calculations of measures of relatedness we use the R package *EconGeo* (Balland, 2017). For standardisation we use the method developed by (Steijn, 2017).

[20] For each region i and each technological class k, we set the sum of relatedness between class k and all other technological classes in which the region i has a revealed technological advantage (RTA) in relation to the sum of technological relatedness between k and all other technologies present in the sample (Europe) (for details, see e.g. Balland et al., 2019; Boschma et al., 2015; Hidalgo et al., 2007). Calculating the average value of relatedness density of technologies that are not yet present in the regional technological portfolio (RTA < 1) for each region, yields a measure (average relatedness density) that is argued to capture the average potential of a region to develop new technological capabilities. For the sake of brevity, we shorten *average relatedness density* to *relatedness density* in this study.

[21] Generally, we use the same metropolitan regions as in our dataset for the calculation of the knowledge complexity index. Due to a lack of data availability or the exclusion of the region from the patent dataset in period 2 (because it produced less patents than the threshold), we need to remove seven metropolitan regions from the dataset of 200 regions (from period 1). These are CH001M, CH002M, CH004MC, CH005M, HR001MC, IT010M, UK517M, see Table A6 for reference.

[22] For the second period (2010-2014), the lead GRP growth is measured from 2015 to 2018 because of data limitations.

models, also regional models (e.g. Piribauer and Crespo Cuaresma, 2016; Sardadvar, 2011), the negative association of GRP with future growth is expected. The number of patents also significantly contributes to future growth. Again, everything else would be unexpected. Most interestingly, our main variable of interest *knowledge complexity* also significantly increases metropolitan economic growth. Note that the KCI shows this positive association in the presence of the number of patents. This shows that knowledge complexity captures a different dimension of knowledge production (than the count of patents) that is also conducive to growth. The remaining control variables *relatedness density* and *diversification* do not seem to notably influence GRP growth.

Since we work with regional data, concerns of spatial dependence in the modelling framework should be addressed (Anselin, 1988; LeSage and Pace, 2009). In order to include spatial effects of any kind in regression analyses, we need to define the neighbourhood structure of metropolitan regions in a so-called *spatial weight matrix* that links each region to its neighbours. We use an inverse distance spatial weight matrix, i.e., every region is connected to every other region with a weight based on the inverse of bilateral distance. Locally robust Lagrange multiplier tests (Anselin et al., 1996) on the baseline model indicate that the spatial error model (SEM) best fits the data. This model allows for a spatially autocorrelated error term. Mainly to serve as robustness checks, we also include a specification that includes a spatial lag of the dependent variable (SAR, model 3) and the combination of both models (SAC, model 4).[23] We see that both the weighted neighbouring residuals and growth rates seem to be positively associated with future metropolitan GRP growth, evidencing a certain spatial dependence in regional economic activities that is to be expected. However, our main independent variable *knowledge complexity* of city-regions does not lose significance in explaining future GRP growth in models 2-4, but is even increasing in terms of relevance relative to *patents*.

We acknowledge that our modelling approach cannot be understood as a precise representation of causal mechanisms of regional economic activities. However, we believe that it serves well as a showcase of the importance of high-quality knowledge production in a regional context, especially when compared to conventional indicators of knowledge production.

4.3. A practical application in a smart specialisation context

In this subsection we turn to a practical application of the concepts presented in this study directly relevant for regional policy. The place-based smart specialisation policy of the European Union (Foray et al., 2009) has been criticised for a lack of operationalisation (Foray et al., 2011; Santoalha, 2019). Balland et al. (2019) introduced an interesting approach to utilise the concept of knowledge complexity to remedy this absence. They do this by combining established literature on regional related diversification (e.g. Boschma, 2017; Breschi et al., 2003; Frenken and Boschma, 2007) with the new concept of knowledge complexity (Balland and Rigby, 2017). For each region a two-dimensional plane can be drawn that relates technological relatedness of a certain technology to its complexity. Policy makers or regional agents can then identify technological domains that are both related to the current regional capabilities as well as more valuable as measured by knowledge complexity. In our practical application we build on this framework but focus on the inter-regional European scale.

In Fig. 3 we show the 211 metropolitan regions of period 2[24] (2010-2014) we analysed previously in Section 4.1 plotted in a two-dimensional space. City-regions are situated according to their relatedness density to not-yet-specialised in technological fields (x-axis) and the potential gain of technological complexity when diversifying into these more or less related technological fields (y-axis). *Potential gain* (in Fig. 3), is defined as the ratio of the average technological complexity (TCI) of technologies not yet present in the region (RTA < 1) and the average TCI of technologies the region already has a revealed technological advantage in.

We can distinguish four quadrants in this framework and are able to classify regions according to these four locations. Regions below the horizontal line are those that are already producing highly complex technological knowledge and arguably should not be incentivised to venture too far from their current specialisations. In this part of the graph we find highly-actively patenting regions like Paris, Eindhoven, London, Frankfurt am Main. Interestingly, Rennes (the top ranked city in terms of KCI) is situated in the bottom-left quadrant, signalling low values in both potential gain and relatedness density. The low relatedness density to other technological fields of Rennes might be concerning as it shows that this region is very specialised in certain rather isolated technological fields, which can be problematic in case these technologies become less valuable (Grabher, 1993). Metropolitan regions that occupy the top-left quadrant have considerable potential in increasing their overall technological knowledge complexity, albeit that these diversification opportunities are at a certain distance in the knowledge space (Breschi et al., 2003; Kogler et al., 2017) and thus associated with higher costs and risks (Balland et al., 2019).

Policy makers might use such a representation to support the conception and design of regional policies. For instance, they might incentivise or provide funding for regions in the top-left quadrant to focus on complex technologies that are most proximate to the regional knowledge base and thus enable them to explore those high-value parts of the knowledge space with less risk. Regions that are located in the top-right quadrant should be able to more easily diversify and increase the complexity of their portfolio because they already possess knowledge that is related to these technologies. Considering their opportunities to strengthen their technological portfolio, these regions should only need guidance in the identification of target technologies but less support in terms of funding or otherwise.

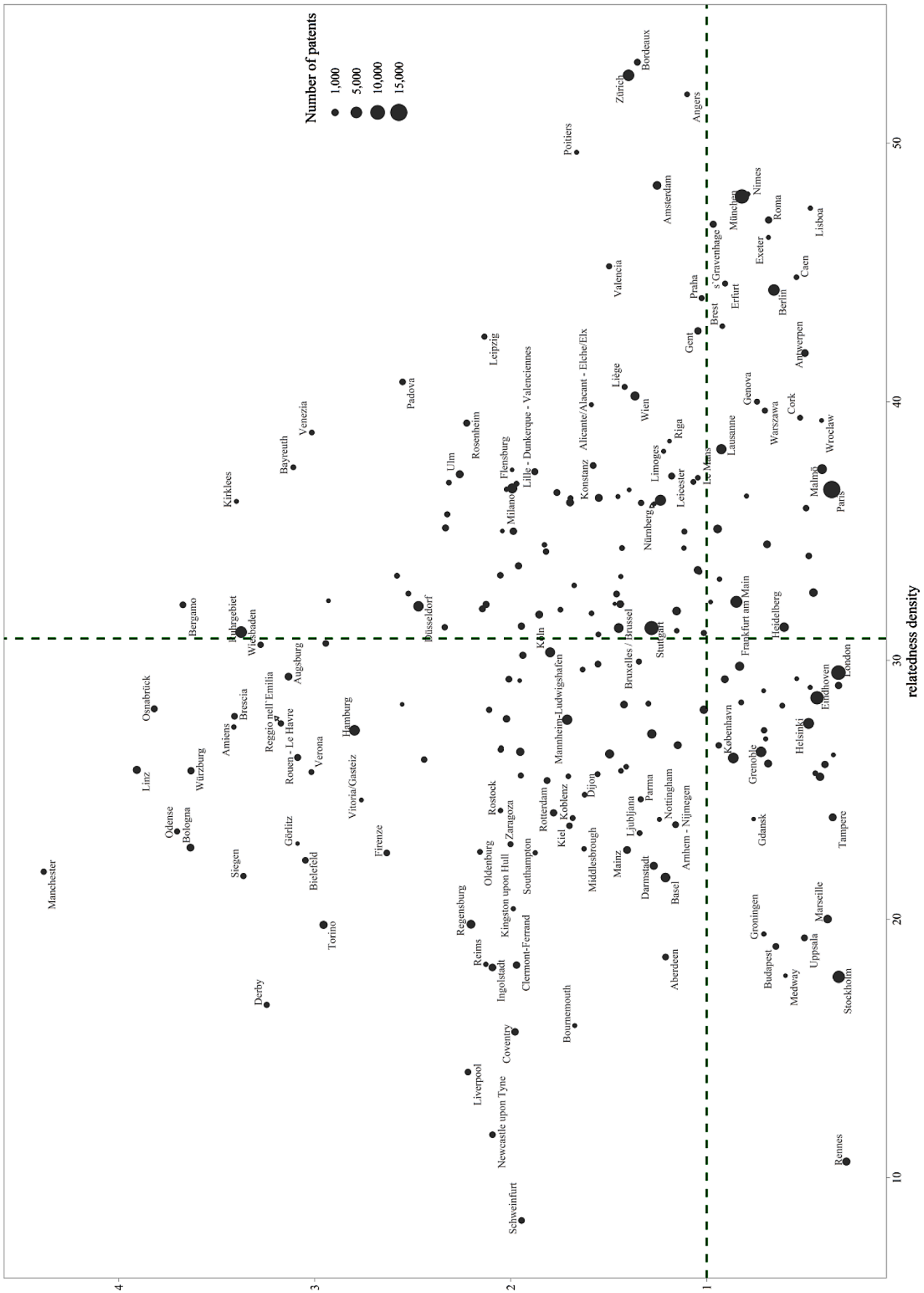
5. Discussion and conclusions

Indicators on knowledge production processes have gained great attention in the recent past, not only in the scientific realm, but also in relation to the increasingly complex design and evaluation of research and innovation policy frameworks at different spatial levels, ranging from the supra-national to the regional level. In the latter context, one main criticism has been that traditional indicators on knowledge production are just able to capture the quantity of new knowledge created, e.g. in a specific spatial entity, while the quality, and even more importantly its value for creating competitive advantage is largely underexplored. Considering for instance the debate on regional smart specialisation policies in Europe (Foray et al., 2009), it is thoroughly emphasised that the capability of regions to diversify into related technological fields by mobilising region-internal and region-external knowledge sources is essential. However, a robust operationalisation of such concepts that is required for the design and evaluation of related policy intervention mechanisms has never reached a sufficiently satisfactory level (see e.g. Balland et al., 2019).

Against this background, a quite recent research stream argues in a regional context that the quality and value of regional knowledge creation might be grasped in a more robust and comprehensive manner by characterising regional diversification patterns, but considering whether the diversified knowledge domains of a region are at the same time scarce as compared to a set of benchmark regions, i.e. how ubiquitous these knowledge domains are (Balland and Rigby, 2017). This concept is recently summarised as knowledge complexity, which is strongly related to the concept of economic complexity introduced by Hidalgo and Hausmann (2009) that looks at the diversified and ubiquitous nature of a

[23] Models 2-4 are estimated with the Maximum Likelihood (ML) estimator using the R package *splm* (Millo and Piras, 2012).

[24] Due to limitations in space we focus on the most recent time period. The corresponding figure for time period 1 is available upon request.



(caption on next page)

Fig. 3. Diversification opportunities of European metropolitan regions.
Notes: Potential gain is the ratio of complexity scores for technologies not yet specialised in compared to technologies in the portfolio of the region. Relatedness density is defined in Section 4.2. The dashed green lines mark the ratio of 1 (y-axis) and the regional average (x-axis). Number of patents refers to citation-adjusted quantity of patents. Note, regions are only labelled if they show a relatedness density of below 25 or above 37. Also, actively patenting regions are named (>2500 patents) as well as regions that show a potential gain ratio of above 3. This is done purely for visualisation reasons.

country's economic structure.

This study follows this related literature by investigating the complex nature of knowledge production analysed at the level of European metropolitan regions by advancing the recently introduced regional knowledge complexity index (KCI). The latter captures regional knowledge endowments in terms of the complexity of technologies in which a region is relatively specialised in. We have applied the index for the first time for a set of 211 European metropolitan regions, using information on regional patenting over the time period 2005-2014. In our empirical setting, we make use of patent citations from extra-regional inventors and accordingly advance existing measures using patents only, accounting for the ability of regions to 'export' knowledge. Moreover, a practical application of the KCI in a smart specialisation policy context is introduced, demonstrating a straightforward and useful addition to the toolkit of policy makers.

The empirical insights of this study are promising, though they are clearly a first step of a longer research agenda. *First*, the results demonstrate that the regional knowledge complexity index (KCI) is indeed a highly interesting concept to capture inherent qualitative differences in regional knowledge production that are not observable by existing measures mainly looking at quantity. *Second*, the regional ranking reveals that there is no obvious relation between knowledge complexity, quantity of patents produced or (non-complex) diversification of regions. Actually, the knowledge complexity index indeed empirically captures a different dimension of knowledge production than the number of patent applications. *Third,* a look at dynamic changes within the last decade of data availability reveals a certain stability of regional knowledge complexity and also technological complexity. Nonetheless, some regions manage to show a remarkable increase in relative ranking between the two periods analysed, while other regions decline significantly. With the given definition of the KCI, such changes are also attributable to increasing or decreasing diversification of regions versus the shifting of knowledge 'exports' to more complex technologies. This is specifically relevant in the current smart specialisation debate, providing a promising entry point for a better operationalisation and monitoring of smart specialisation policies. *Fourth,* the analysis of the spatial distribution of the KCI reveals a certain geographical logic in knowledge creation activities across the European territory. For instance, traditionally industrial central European regions are often labelled relatively non-complex in terms of knowledge production, while new regional players, usually not on top in the regional patent ranking rank quite prominently. *Fifth,* knowledge complexity is shown to be associated with future regional economic growth, in addition to overall knowledge production activities of the region. *Finally,* a practical application of regional knowledge complexity measures is introduced to classify regions according to their potential to diversify into promising technological fields and their expected ease of realising this endeavour. Applications of such kind may be an extremely valuable addition for benchmarking regions in the context of smart specialisation policies, supported by regional, national or European policy interventions (see e.g. Uhlbach et al., 2017).

Clearly, to become a usable and robust policy support and evaluation framework in the future, the approach presented here needs further advancement and refinement. This is related to the main limitations of the index, in particular its sensitivity on the underlying empirical operationalisation of technological fields. In addition, we face the well-known limitations of patent indicators per se, i.e. patenting is always a strategic decision and characterised by a differing patenting propensity across different fields. The latter leads to a situation where, in contrast to the production and exporting of economic goods, also lagging regions – though having a small patent portfolio in terms of magnitude – assign patents in more complex technologies than to be expected. Though we have tried to reduce this bias by accounting for the regional knowledge 'exporting' capability (in terms of patent citations), this issue needs thorough re-consideration, robustness testing and potential adaptations.

Against this background, future research should focus on the following main points: *First*, the measure needs – as mentioned above – more robustness and sensitivity checks, and a systematic benchmarking to alternative conceptualisations of knowledge complexity. One promising alternative may be to define the complexity of a technology by approaches grasping the re-combinatorial rareness of a technology in a search process over time, rather than just looking at the actual ubiquitous nature of a specific technology (see e.g. Fleming and Sorenson, 2001; Sorenson et al., 2006). Another interesting avenue would be to identify technological complexity as structural complexity of the combinatorial network of technology (sub)classes (Broekel, 2019). *Second*, given the spatial clustering tendencies of complex knowledge, it will be crucial to identify factors that facilitate and hinder regional diversification into more complex technologies. This will also provide further important insights into the mobility of complex knowledge in geographical space. *Third*, and most importantly, future work should focus on characterising the relationships – by means of dynamic spatial econometric models – between knowledge complexity and regional total-factor productivity, assuming that regions with a higher knowledge complexity show higher productivity gains in the long term. Here, mediating factors of the regional innovation capacity as well as the effective time horizon of productivity effects of knowledge complexity will have to be taken very seriously.

CRediT authorship contribution statement

Nico Pintar: Conceptualization, Methodology, Software, Validation, Formal analysis, Data curation, Writing - original draft, Writing - review & editing, Visualization. **Thomas Scherngell:** Conceptualization, Methodology, Validation, Writing - original draft, Writing - review & editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Figs. A1–A2, Tables A1–A6

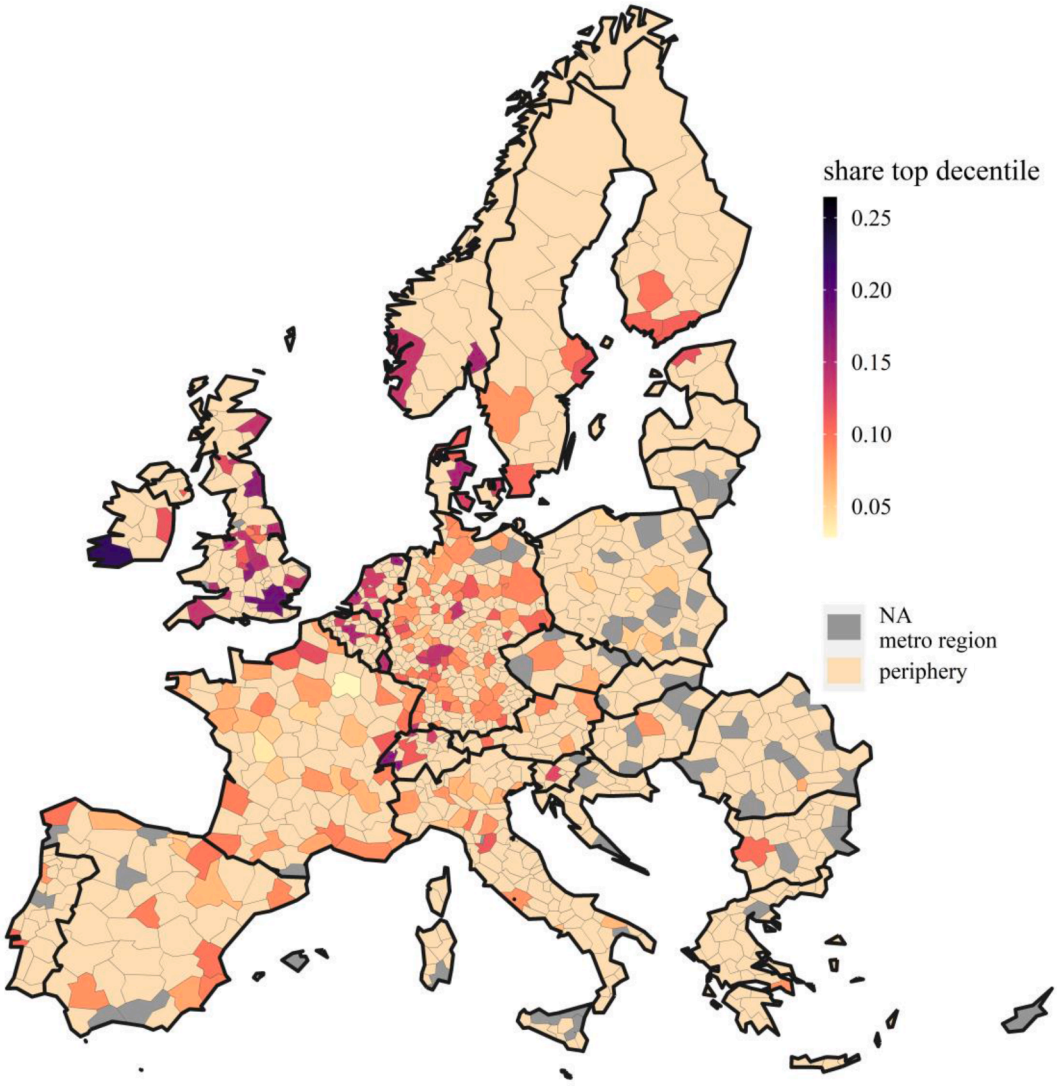


Fig. A1. Spatial distribution of extra-regional patent citations received, 2010-2014.
Notes: Share top decentile refers to the relative share of patent applications of a region that rank in the top decentile in terms of citations received within the same cohort (year, technology).

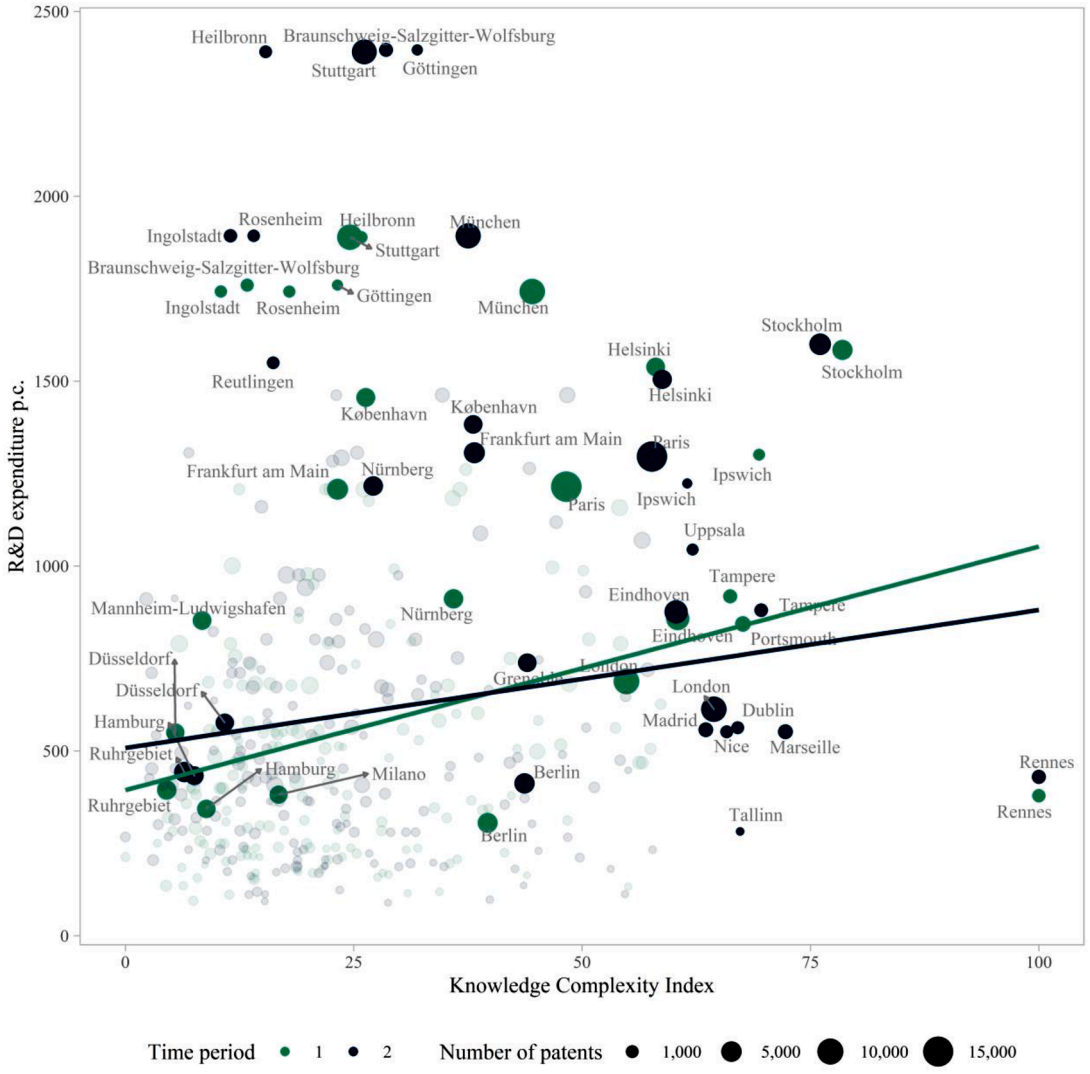


Fig. A2. Comparison of KCI vs. R&D expenditure per capita, both periods.
Notes: Regions are pictured for both periods under investigation and coloured accordingly. A simple linear fit (not taking into account the number of patents or size of the circle) is shown for each period. R&D expenditure data is sourced from Eurostat and given in per capita pps. R&D data is based on NUTS2 data and approximated for metropolitan regions, assuming same R&D expenditure in NUTS3 regions within respective NUTS2 regions. Number of patents refers to citation-adjusted quantity of patents. Note, regions are only labelled if they have KCI>60 and or R&D expenditure of more than 1500. Also, actively patenting regions are named (>3500 patents). All other regions are not named and shown transparent. This is done purely for visualisation reasons.

Table A1
Top 25 metropolitan regions in terms of knowledge complexity, 2005–2009.

Metro code	Name	No. of patents	KCI	KCI rank	Div.
FR013M	Rennes	1204.2	100.0	1	5
SE001MC	Stockholm	4984.0	78.5	2	8
UK569M	Ipswich	693.0	69.4	3	9
UK023M	Portsmouth	2099.3	67.6	4	12
FI002M	Tampere	1455.5	66.2	5	9
NL005M	Eindhoven	7616.6	60.5	6	9
FI003M	Turku	675.9	58.6	7	9
FI001MC	Helsinki	3978.6	58.0	8	8
IE001MC	Dublin	826.8	57.5	9	13
FR205M	Nice	1138.7	56.6	10	10
FR203M	Marseille	1302.4	56.2	11	11
IT010M	Catania	144.1	55.1	12	9
UK001MC	London	9957.7	54.9	13	17
DE507M	Aachen	2053.1	54.2	14	13
SE003M	Malmö	2950.0	54.1	15	15
UK007M	Edinburgh	649.4	53.7	16	14
UK011M	Bristol	1354.0	50.7	17	12
SE006M	Uppsala	661.9	50.1	18	12
FR026M	Grenoble	2810.2	48.6	19	14
DK004M	Aalborg	401.5	48.4	20	11
FR001MC	Paris	15693.7	48.3	21	14
IT001MC	Roma	1093.4	47.7	22	12
FR004M	Toulouse	1409.7	46.8	23	9
DE542M	Hildesheim	437.7	45.2	24	11
AT001MC	Wien	2425.6	45.1	25	16

Notes: Patents are citation-adjusted and fractionally counted. KCI score is re-scaled to values between 100 and zero.

Table A2
Summary statistics, 2005–2014.

Statistic	N	Min	Mean	Median	Max	SD
GRP growth	386	-5.3	1.5	1.6	10.1	1.4
initial GRP	386	6433.1	43503.9	22820.8	745082.2	73363.4
KCI	386	-1.5	0.0	-0.3	4.6	1.0
No. of patents	386	89.6	1196.5	592.7	15693.7	1872.5
Relatedness Density	386	7.9	30.2	30.3	53.1	7.9
Diversification	386	3	12	12	19	2.8
TCI	70	-1.2	0.0	-0.5	2.6	1.0
Ubiquity	70	35.0	70.8	72.0	106.0	16.9
No. of patents	70	401.0	6908.7	5832.4	16845.0	4061.0

Notes: Mind that the upper part of the table corresponds regional data used for the model in Table 2. The lower part of the table refers to technological fields. Patents are citation-adjusted and fractionally counted. Complexity scores are normalised.

Table A3
Correlation table, 2005–2014.

	GRP gr.	init. GRP	KCI	Patents	Relat. Dens.
GRP growth					
initial GRP	0.041				
KCI	0.178	0.261			
No. of patents	0.109	0.750	0.293		
Relatedness Density	0.102	0.100	0.039	0.046	
Diversification	0.079	0.200	0.047	0.140	0.888
	TCI	**Patents**			
TCI					
No. of patents	0.008				
Ubiquity	-0.726	0.082			

Notes: Mind that the upper part of the table corresponds regional data used for the model in Table 2. The lower part of the table refers to technological fields. Patents are citation-adjusted and fractionally counted. Complexity scores are normalised.

Table A4
Technological fields in terms of technological complexity, 2010–2014.

Field	Description	No. of patents	TCI	TCI rank	Ubi.
4	Digital communication	14562.6	100.0	1	37
6	Computer technology	11159.1	83.6	2	52
3	Telecommunications	4766.8	79.1	3	53
2	Audio-visual technology	5120.0	75.4	4	39
7	IT methods for management	2003.3	74.2	5	53
5	Basic communication processes	1684.4	62.7	6	51
9	Optics	3285.2	46.3	7	70
15	Biotechnology	7279.7	44.9	8	76
11	Analysis of biological materials	2347.7	42.5	9	80
16	Pharmaceuticals	9256.1	39.5	10	84
10	Measurement	13025.5	39.1	11	68
22	Micro-structural and nano-technology	442.8	34.3	12	62
12	Control	4279.5	34.1	13	77
13	Medical technology	14519.5	33.3	14	80
8	Semiconductors	4012.9	32.8	15	64
24	Environmental technology	3714.4	29.5	16	81
18	Food chemistry	2435.9	25.9	17	77
14	Organic fine chemistry	8165.3	24.6	18	67
23	Chemical engineering	6354.9	18.1	19	96
33	Furniture, games	5059.1	15.7	20	77
19	Basic materials chemistry	7053.3	15.7	21	64
32	Transport	15647.6	15.1	22	64
20	Materials, metallurgy	4626.0	13.0	23	93
17	Macromolecular chemistry, polymers	5191.5	12.4	24	72
28	Textile and paper machines	3312.2	12.3	25	78
35	Civil engineering	9620.4	8.8	26	99
1	Electrical machinery, apparatus, energy	16845.0	8.7	27	76
30	Thermal processes and apparatus	4755.4	8.4	28	98
29	Other special machines	8404.2	8.0	29	102
34	Other consumer goods	5326.8	7.9	30	57
25	Handling	8452.3	7.1	31	89
21	Surface technology, coating	3497.7	5.8	32	84
27	Engines, pumps, turbines	10934.7	5.5	33	68
26	Machine tools	5918.2	2.1	34	79
31	Mechanical elements	9527.0	0.0	35	95

Notes: Patents are citation-adjusted and fractionally counted. TCI score is re-scaled to values between 100 and zero.

Table A5
Technological fields in terms of technological complexity, 2005–2009.

Field	Description	No. of patents	TCI	TCI rank	Ubiquity
4	Digital communication	13789.0	100.0	1	35
3	Telecommunications	5746.6	97.4	2	42
6	Computer technology	10746.3	87.5	3	48
2	Audio-visual technology	6764.2	76.0	4	37
5	Basic communication processes	1949.6	69.8	5	58
7	IT methods for management	1586.7	63.0	6	50
9	Optics	3784.5	49.3	7	52
22	Micro-structural and nano-technology	401.0	47.6	8	48
8	Semiconductors	4384.5	46.3	9	50
13	Medical technology	13268.8	43.3	10	75
10	Measurement	11907.3	36.8	11	77
11	Analysis of biological materials	2517.5	33.5	12	72
12	Control	4164.3	31.6	13	76
16	Pharmaceuticals	12501.7	25.7	14	79
1	Electrical machinery, apparatus, energy	13379.2	25.5	15	84
15	Biotechnology	7653.8	22.6	16	76
32	Transport	15473.6	15.5	17	74
14	Organic fine chemistry	10724.2	14.2	18	63
18	Food chemistry	2612.6	13.9	19	67
27	Engines, pumps, turbines	9279.0	13.3	20	67
34	Other consumer goods	5011.4	12.1	21	61
24	Environmental technology	3414.3	12.0	22	66
33	Furniture, games	4766.0	11.6	23	76
21	Surface technology, coating	3497.4	10.4	24	67
28	Textile and paper machines	4613.3	8.8	25	70
23	Chemical engineering	6330.7	7.1	26	86
25	Handling	8474.2	6.0	27	92
30	Thermal processes and apparatus	4140.4	5.9	28	85
35	Civil engineering	8857.5	5.6	29	106
26	Machine tools	6106.7	5.3	30	69
31	Mechanical elements	9336.1	4.8	31	85
20	Materials, metallurgy	3933.8	3.5	32	86
19	Basic materials chemistry	7657.5	2.6	33	49
17	Macromolecular chemistry, polymers	5101.9	1.9	34	67
29	Other special machines	7149.8	0.0	35	100

Notes: Patents are citation-adjusted and fractionally counted. TCI score is re-scaled to values between 100 and zero.

Table A6
List of all regions in the dataset.

Metro code	Name	KCI rank 05-09	KCI rank 10-14
AT001MC	Wien	25	90
AT002M	Graz	100	105
AT003M	Linz	174	210
AT004M	Salzburg	77	158
AT005M	Innsbruck	54	69
BE001MC	Bruxelles / Brussel	155	96
BE002M	Antwerpen	30	25
BE003M	Gent	93	62
BE004M	Charleroi	200	70
BE005M	Liège	106	93
BG001MC	Sofia	excluded	58
BG002M	Plovdiv	excluded	excluded
BG003M	Varna	excluded	excluded
BG004M	Burgas	excluded	excluded
CH001M	Zürich	81	85
CH002M	Genève	97	54
CH003M	Basel	103	77
CH004MC	Bern	59	60
CH005M	Lausanne	50	53
CY001MC	Lefkosia	excluded	excluded
CZ001MC	Praha	40	61
CZ002M	Brno	92	134
CZ003M	Ostrava	excluded	excluded
CZ004M	Plzen	excluded	excluded
CZ007M	Liberec	excluded	excluded
DE001MC	Berlin	33	37
DE002M	Hamburg	175	189
DE003M	München	26	49
DE004M	Köln	192	129
DE005M	Frankfurt am Main	85	47
DE007M	Stuttgart	76	83
DE008M	Leipzig	128	143
DE009M	Dresden	90	132
DE011M	Düsseldorf	193	177
DE012M	Bremen	177	161
DE013M	Hannover	47	73
DE014M	Nürnberg	41	79
DE017M	Bielefeld	154	201
DE018M	Halle an der Saale	143	101
DE019M	Magdeburg	144	92
DE020M	Wiesbaden	150	190
DE021M	Göttingen	83	63
DE025M	Darmstadt	88	87
DE027M	Freiburg im Breisgau	44	65
DE028M	Regensburg	51	178
DE031M	Schwerin	excluded	excluded
DE032M	Erfurt	36	51
DE033M	Augsburg	184	196
DE034M	Bonn	157	121
DE035M	Karlsruhe	38	56
DE036M	Mönchengladbach	117	142
DE037M	Mainz	138	99
DE038M	Ruhrgebiet	196	193
DE039M	Kiel	66	133
DE040M	Saarbrücken	127	153
DE042M	Koblenz	182	130
DE043M	Rostock	84	163
DE044M	Kaiserslautern	74	107
DE045M	Iserlohn	189	188
DE052M	Flensburg	excluded	140
DE054M	Konstanz	114	114
DE057M	Gießen	118	141
DE059M	Bayreuth	160	183
DE061M	Aschaffenburg	173	167
DE064M	Neubrandenburg	excluded	excluded
DE069M	Rosenheim	115	155
DE073M	Offenburg	146	144
DE074M	Görlitz	excluded	199
DE077M	Schweinfurt	98	173
DE079M	Wetzlar	82	88
DE083M	Braunschweig-Salzgitter-Wolfsburg	142	74
DE084M	Mannheim-Ludwigshafen	179	118
DE504M	Münster	172	154
DE507M	Aachen	14	112
DE510M	Lübeck	116	116
DE513M	Kassel	78	164
DE517M	Osnabrück	197	206
DE520M	Oldenburg	190	172
DE522M	Heidelberg	70	28
DE523M	Paderborn	87	115
DE524M	Würzburg	169	207
DE527M	Bremerhaven	149	182
DE529M	Heilbronn	71	146
DE532M	Ulm	104	149
DE533M	Pforzheim	94	102
DE534M	Ingolstadt	165	174
DE537M	Reutlingen	68	138
DE540M	Siegen	181	204
DE542M	Hildesheim	24	44
DE544M	Zwickau	72	104
DE546M	Wuppertal	170	175
DK001MC	København	69	48
DK002M	Århus	139	156
DK003M	Odense	123	209
DK004M	Aalborg	20	59
EE001MC	Tallinn	excluded	5
EL001MC	Athina	39	26
EL002M	Thessaloniki	excluded	excluded
ES001MC	Madrid	35	9
ES002M	Barcelona	111	86
ES003M	Valencia	148	100
ES004M	Sevilla	112	55
ES005M	Zaragoza	86	162
ES006M	Málaga - Marbella	excluded	excluded
ES007M	Murcia - Cartagena	excluded	157
ES008M	Las Palmas	excluded	excluded
ES009M	Valladolid	excluded	excluded
ES010M	Palma de Mallorca	excluded	excluded
ES012M	Vitoria/Gasteiz	156	192
ES013M	Oviedo - Gijón	176	110
ES014M	Pamplona/Iruña	125	170
ES015M	Santander	excluded	excluded
ES019M	Bilbao	122	128
ES021M	Alicante/Alacant - Elche/Elx	137	109
ES022M	Vigo	excluded	excluded
ES025M	Santa Cruz de Tenerife	excluded	excluded
ES026M	A Coruña	excluded	22
ES501M	Granada	excluded	excluded
FI001MC	Helsinki	8	13
FI002M	Tampere	5	4
FI003M	Turku	7	19
FR001MC	Paris	21	15
FR003M	Lyon	134	108
FR004M	Toulouse	23	30
FR006M	Strasbourg	164	145
FR007M	Bordeaux	65	81
FR008M	Nantes	102	89
FR009M	Lille - Dunkerque - Valenciennes	135	125
FR010M	Montpellier	52	39
FR011M	Saint-Etienne	159	171
FR013M	Rennes	1	1
FR014M	Amiens	199	202
FR015M	Rouen - Le Havre	195	198
FR016M	Nancy	178	127
FR018M	Reims	191	176
FR019M	Orléans	180	180
FR020M	Dijon	141	123
FR021M	Poitiers	110	111
FR022M	Clermont-Ferrand	188	168
FR023M	Caen	31	27
FR024M	Limoges	95	78
FR025M	Besanþon	99	106
FR026M	Grenoble	19	35
FR035M	Tours	32	117
FR036M	Angers	75	67
FR037M	Brest	29	52
FR038M	Le Mans	132	64

(*continued on next page*)

Table A6 (*continued*)

Metro code	Name	KCI rank 05-09	KCI rank 10-14
FR040M	Mulhouse	168	126
FR043M	Perpignan	excluded	excluded
FR044M	Nimes	151	46
FR045M	Pau	162	165
FR048M	Annecy (FR)/Genève (CH)	126	159
FR203M	Marseille	11	3
FR205M	Nice	10	7
HR001MC	Grad Zagreb	27	excluded
HR005M	Split	excluded	excluded
HU001MC	Budapest	34	23
HU002M	Miskolc	excluded	excluded
HU004M	Pécs	excluded	excluded
HU005M	Debrecen	excluded	excluded
HU009M	Székesfehérvár	excluded	excluded
IE001MC	Dublin	9	6
IE002M	Cork	28	24
IT001MC	Roma	22	41
IT002M	Milano	119	137
IT003M	Napoli	43	68
IT004M	Torino	171	195
IT005M	Palermo	excluded	excluded
IT006M	Genova	45	42
IT007M	Firenze	130	187
IT008M	Bari	46	98
IT009M	Bologna	185	208
IT010M	Catania	12	excluded
IT011M	Venezia	163	185
IT012M	Verona	167	194
IT022M	Taranto	excluded	excluded
IT027M	Cagliari	excluded	excluded
IT028M	Padova	105	166
IT029M	Brescia	198	200
IT501M	Messina	excluded	excluded
IT502M	Prato	excluded	excluded
IT503M	Parma	187	94
IT505M	Reggio nell`Emilia	166	197
IT511M	Bergamo	183	203
LT001MC	Vilnius	excluded	excluded
LT002M	Kaunas	excluded	excluded
LU001MC	Luxembourg	58	184
LV001MC	Riga	excluded	71
MT001MC	Valletta	excluded	excluded
NL001M	s`Gravenhage	57	57
NL002MC	Amsterdam	109	76
NL003M	Rotterdam	113	136
NL004M	Utrecht	152	103
NL005M	Eindhoven	6	12
NL006M	Tilburg	108	120
NL007M	Groningen	48	31
NL008M	Enschede	107	151
NL009M	Arnhem - Nijmegen	121	72
NO001MC	Oslo	37	34
NO002M	Bergen	145	139
PL001MC	Warszawa	53	36
PL002M	Lódz	excluded	43
PL003M	Kraków	excluded	113
PL004M	Wroclaw	excluded	20
PL005M	Poznan	excluded	147
PL006M	Gdansk	excluded	38
PL007M	Szczecin	excluded	excluded
PL008M	Bydgoszcz - Torún	excluded	excluded
PL009M	Lublin	excluded	excluded
PL010M	Katowice	excluded	148
PL011M	Bialystok	excluded	excluded
PL012M	Kielce	excluded	excluded
PL014M	Olsztyn	excluded	excluded
PL015M	Rzeszów	excluded	excluded
PL016M	Opole	excluded	excluded
PL024M	Czestochowa	excluded	excluded
PL025M	Radom	excluded	excluded
PL506M	Bielsko-Biala	excluded	excluded
PL514M	Tarnów	excluded	excluded
PT001MC	Lisboa	61	29
PT002M	Porto	136	80
PT005M	Coimbra	excluded	excluded
RO001MC	Bucuresti	excluded	32
RO002M	Cluj-Napoca	excluded	excluded
RO003M	Timisoara	excluded	excluded
RO004M	Craiova	excluded	excluded
RO501M	Constanta	excluded	excluded
RO502M	Iasi	excluded	excluded
RO503M	Galati	excluded	excluded
RO504M	Brasov	excluded	excluded
RO505M	Ploiesti	excluded	excluded
SE001MC	Stockholm	2	2
SE002M	Göteborg	42	45
SE003M	Malmö	15	17
SE006M	Uppsala	18	10
SI001MC	Ljubljana	63	95
SI002M	Maribor	excluded	excluded
SK001MC	Bratislava	excluded	excluded
SK002M	KoÜice	excluded	excluded
UK001MC	London	13	8
UK002M	West Midlands urban area	140	131
UK003M	Leeds	101	119
UK004M	Glasgow	73	97
UK005M	Bradford	64	14
UK006M	Liverpool	147	181
UK007M	Edinburgh	16	21
UK008M	Manchester	194	211
UK009M	Cardiff	89	66
UK010M	Sheffield	80	122
UK011M	Bristol	17	50
UK012M	Belfast	56	33
UK013M	Newcastle upon Tyne	120	179
UK014M	Leicester	91	75
UK016M	Aberdeen	129	82
UK018M	Exeter	67	40
UK023M	Portsmouth	4	16
UK025M	Coventry	49	169
UK026M	Kingston upon Hull	158	160
UK027M	Stoke-on-Trent	186	152
UK029M	Nottingham	96	84
UK501M	Kirklees	133	191
UK506M	Doncaster	131	186
UK510M	Sunderland	excluded	excluded
UK513M	Medway	79	18
UK515M	Brighton and Hove	excluded	excluded
UK517M	Swansea	55	excluded
UK518M	Derby	153	205
UK520M	Southampton	60	150
UK539M	Bournemouth	62	135
UK553M	Blackburn - Blackpool - Preston	excluded	excluded
UK559M	Middlesbrough	161	124
UK568M	Cheshire West and Chester	124	91
UK569M	Ipswich	3	11

Notes: Metropolitan regions that were excluded because they did not produce enough patents are marked with "excluded". See footnote 9 for details about the used metropolitan regions as defined by EUROSTAT.

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